

Re-analysis of GEO GSE54825: transcriptional response of engineered *Saccharomyces cerevisiae* to cellobiose versus glucose

A reproducible pipeline with a quantitative comparison to the original publication

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2026-09-14

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1 Background

Saccharomyces cerevisiae cannot metabolise cellobiose. Engineering a heterologous cellobiose utilisation pathway — a cellodextrin transporter (cdt-1) and an intracellular β -glucosidase (gh1-1) from *Neurospora crassa* — enables growth on this sugar, but fermentation remains slower than on glucose (Galazka et al. 2010). The original study of the data set re-analysed here (GEO [GSE54825](#); (Lin et al. 2014)) profiled the transcriptome of the engineered strain on cellobiose versus glucose under anaerobic conditions and reported 519 differentially expressed genes, mitochondrial activation, reduced amino-acid and thiamine biosynthesis, and two engineering leads (SUT1 overexpression, HAP4 deletion).

That study was analysed with a commercial platform (CLC Genomics Workbench 6.5), a private reference assembly, and a statistical framework (Baggerley’s test) that is no longer standard. This report re-analyses the same raw reads with a modern, fully scripted pipeline and asks a question the original could not answer: **how much of the original result is reproducible, and what is not assessable at all?**

2 Research question

What is the global transcriptional difference between cellobiose- and glucose-grown engineered *S. cerevisiae* under anaerobic exponential growth; which biological processes are involved; and which observations from the original study can be quantitatively reproduced with an independent pipeline?

3 Dataset

The seven input files (paper and supplements of both studies) are archived read-only with md5 checksums; the sequencing data are public and were verified against the md5 values published by ENA.

sample_id	gsm	run	condition	replicate	layout	platform
JCYL001B	GSM1324496	SRR1166442	glucose	1	SE	GAI
JCYL002B	GSM1324497	SRR1166443	glucose	2	SE	GAI
JCYL003B	GSM1324498	SRR1166444	glucose	3	SE	GAI
JCYL001D	GSM1324499	SRR1166445	cellobiose	1	SE	GAI
JCYL002D	GSM1324500	SRR1166446	cellobiose	2	SE	GAI
JCYL003D	GSM1324501	SRR1166447	cellobiose	3	SE	GAI

run	sample_id	condition	file	bytes	md5_self	md5_ena	status
SRR1166442	JCYL001B	glucose	SRR1166442.fastq.gz	1957980073	68e674b8a20e	68e674b8a20e	PASS
SRR1166443	JCYL002B	glucose	SRR1166443.fastq.gz	2168810515	dcc91f003612	dcc91f003612	PASS
SRR1166444	JCYL003B	glucose	SRR1166444.fastq.gz	1601609705	f58e37e44319	f58e37e44319	PASS
SRR1166445	JCYL001D	cellobiose	SRR1166445.fastq.gz	2189433744	a201447b2dda	a201447b2dda	PASS
SRR1166446	JCYL002D	cellobiose	SRR1166446.fastq.gz	1839997905	028b8c4edc3b	028b8c4edc3b	PASS
SRR1166447	JCYL003D	cellobiose	SRR1166447.fastq.gz	1586975675	cbb987dbfaa3	cbb987dbfaa3	PASS

4 Experimental design

Two conditions $\times$ three biological replicates, a single batch, anaerobic exponential phase (80 g/L starting sugar, harvested at $\approx$ 50 g/L residual sugar). The original study used an unpaired two-group comparison; replicate numbering (JCYL001B/001D, ...) hints at pairing but this cannot be verified from public metadata. The primary model is therefore unpaired ($\sim$ condition) and a paired model ($\sim$ replicate + condition) is run as a sensitivity analysis (see the *Differential expression* section). No spike-in controls and no independent batch are available.

5 Methods

Step	Tool, version and settings
Raw data	ENA FTP download; md5 verified against ENA-published values
Read QC	FastQC 0.12.1; fastp 1.3.6 (light trimming, length $\geq$ 36 bp); MultiQC 1.35
Reference (host)	RefSeq GCF_000146045.2 (S288C R64-1-1, 16 chromosomes + mitochondrion); transcripts extracted with gffread 0.12.9; annotation from the SGD GAF (release R64-5-1)
Reference (trans-genes)	Host-vector-insert sequences adjudicated from the reads themselves (see the <i>Quantification</i> section)
Quantification	Salmon 2.7.0, decoy-aware index (k = 31; k = 25 as sensitivity), $-\text{libType A}$ $-\text{gcBias}$ $-\text{seqBias}$, fragment-length prior set to the observed read length (49 bp)
Gene-level matrix	tximport 1.34.0, countsFromAbundance = lengthScaledTPM
Differential expression	DESeq2 1.46.0; primary $\sim$ condition, sensitivity $\sim$ replicate + condition; LFC shrinkage (apeglm) for display only
Multiple testing	Benjamini-Hochberg
Functional enrichment	clusterProfiler 4.14.0 (ORA + GSEA) with self-built TERM2GENE tables; KEGG (sce) cached
Report	Quarto 1.9.38 (HTML; PDF via LaTeX/xelatex)

Thresholds. Primary DEGs are BH-adjusted $p < 0.05$ and $|\log_2\text{FC}| > 1$. A relaxed ($p < 0.1$) and a stringent tier ($p < 0.01$ and $|\log_2\text{FC}| > 1.5$) are also reported, plus a tier matched to

the original study ($\text{FDR} \leq 0.001$ and $|\log_2\text{FC}| \geq 1$). **All thresholding uses unshrunk (MLE) $\log_2\text{FC}$ values**; apeglm-shrunk values are used only for ranking and plots. The enrichment universe is the set of expressed genes (count ≥ 10 in at least three samples): 6049 genes.

6 Quality control

The 2014 data are single-end 50 bp reads from a Genome Analyzer II. FastQC reported Sanger/Illumina 1.9 encoding, so no Phred64 handling was required. Light trimming retained 95.1% of 238.6 million reads and raised the Q30 fraction from 0.88–0.93 to 0.93–0.96; adapter content is negligible (≈ 150 reads per sample after trimming, 0.01%).

Read fate. Reads were attributed in three tiers: (i) end-to-end alignment to the host genome plus transgenes (94.5–95.5%), (ii) local-mode re-alignment of the remainder (recovering 19.9–22.8% of the unmapped fraction, attributable to the strict end-to-end mode and to read errors in these older data), and (iii) a diagnostic reference containing the pRS426 backbone, the native 2 μm plasmid and TruSeq adapters (a further 35.6–40.4% of the residual). A small, quantified fraction (2.2–3.0% of the library) remains unexplained; candidate explanations are listed in the *Limitations* section. Multi-mapping is 23.0–24.9%, driven by Ty elements and the rDNA repeat, and the rDNA locus is the most deeply covered region of the reference (RDN25-1, mean depth 23 $\times$).

Plasmid copy number. Reads from the engineered plasmid backbone are 1.65 $\times$ more abundant on cellobiose (10,385 $\rightarrow$ 17,188 reads per million), whereas reads from the native 2 μm plasmid are unchanged (3,358 $\rightarrow$ 3,012). This is independent evidence that any copy-number effect is specific to the engineered plasmid.

7 Quantification

The host reference and annotation were frozen (RefSeq GCF_000146045.2, SGD annotation release R64-5-1, all files md5-recorded). Two properties of this data set required specific handling.

The transgenes are not what the literature suggests. The RNA-seq samples carry pRS426-BT, whose inserts derive from *N. crassa* cDNA. Reads-derived consensus sequences match the public CDS of *cdt-1* (NCU00801) and *gh1-1* (NCU00130) base-for-base (0 differences). The codon-optimised *gh1-1a* published in the supplement (Figure S8) is **absent** from the samples (2 reads in 6 million) and differs from the native CDS at 25% of positions: using it as a reference would have silently destroyed transgene quantification. The C-terminal tag is a **superfolder GFP**: the consensus differs from the humanised EGFP CDS at only 7 nucleotides, and its translation carries all six superfolder substitutions (S30R/Y39N/N105T/Y145F/I171V/A206V) on an EGFP backbone. The *cdt-1*-tag junction was read directly from soft-clipped reads: $\cdots\text{GTTGCT} + \text{ATCGAT (Clal)} + \text{GGTAGTGGTAGT (Gly-Ser-Gly-Ser)} + \text{GTGAGC}\cdots$, consistent with the cloning strategy described in the earlier work (Galazka et al. 2010). Because part of the transgene reference is derived from the sample reads, transgene results are reported **as ratios only**.

Mitochondrially encoded genes must be measured, not assumed. The RefSeq RNA FASTA contains no mitochondrial protein-coding mRNA, so transcripts were extracted from the GFF with gffread (6480 host transcripts, including 19 mitochondrial mRNAs) plus the two transgenes.

Salmon with a decoy-aware index mapped 80.0–83.5% of reads ($k = 31$); library type was inferred as unstranded (**U**) in all twelve runs, consistent with TruSeq v1 chemistry. Quantification was validated in three independent ways: (i) TPM versus the original RPKM correlates with Spearman = 0.880 over 5796 genes; (ii) adding or removing the genome decoy changes TPMs by a median of 0.027 log2 units (17 of 5960 genes exceed 1); and (iii) two calibration genes reproduce the original values almost exactly (URA3 ratio 1.364 versus 1.354; PGK1 ratio 0.749 versus 0.745).

Reported EffectiveLength values are anomalous in Salmon 2.7.0 (Rust port): host transcripts receive a median effective length of $0.465 \times \text{length}$, and the transgene entries receive values above $10 \times \text{length}$. This does not affect the conclusions above; the checks in (i)–(iii) and the decoy comparison were performed specifically to bound the risk, and all transgene statements are based on ratios.

8 Sample-level analysis

item	value
n_genes_used	6049
PC1_var_pct	93.4
PC2_var_pct	2.3
within_glucose_spearman_min	0.9961
within_glucose_spearman_max	0.9974
within_cellobiose_spearman_min	0.9944
within_cellobiose_spearman_max	0.9977
between_condition_spearman_min	0.8855
between_condition_spearman_max	0.8954
separation_verdict	conditions separated on PC1

Conditions separate on PC1 (93.4% of variance) and replicates within a condition are extremely concordant (Spearman ρ 0.994–0.998) compared with the between-condition range (0.886–0.895), supporting H_0 .

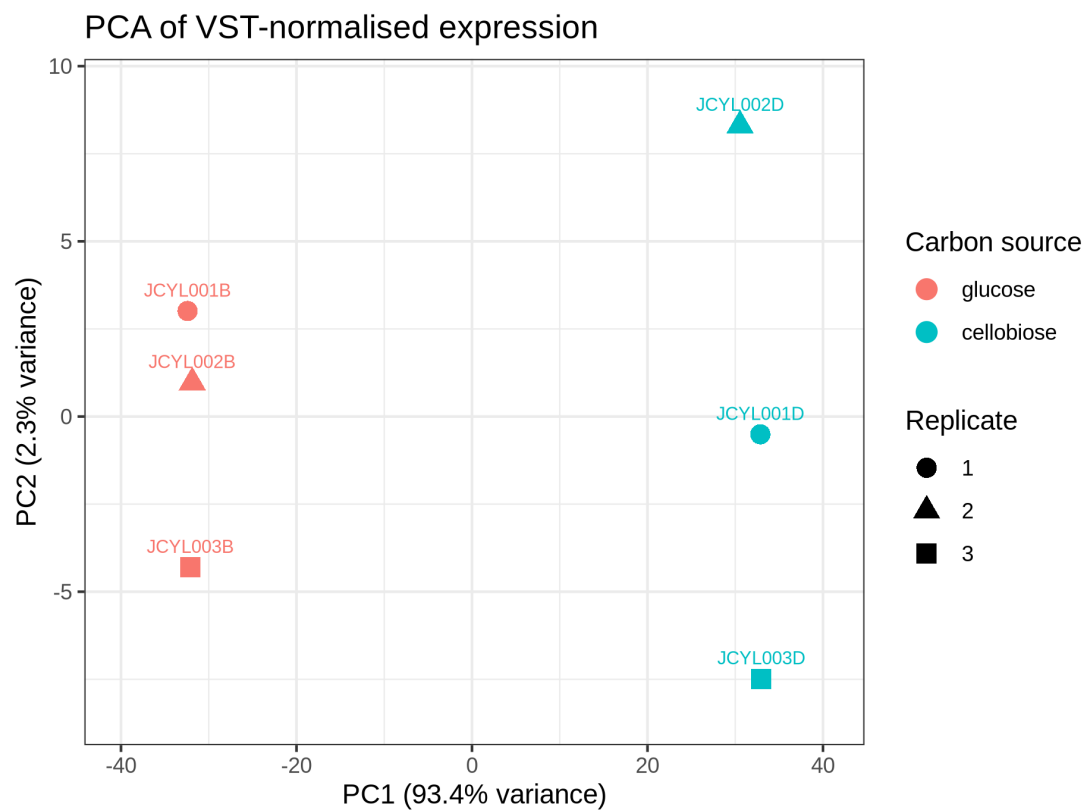


Figure 1: PCA of VST-normalised expression. Conditions separate on PC1 (93.4% of variance); within-condition replicates are tightly clustered (Spearman rho 0.994-0.998).

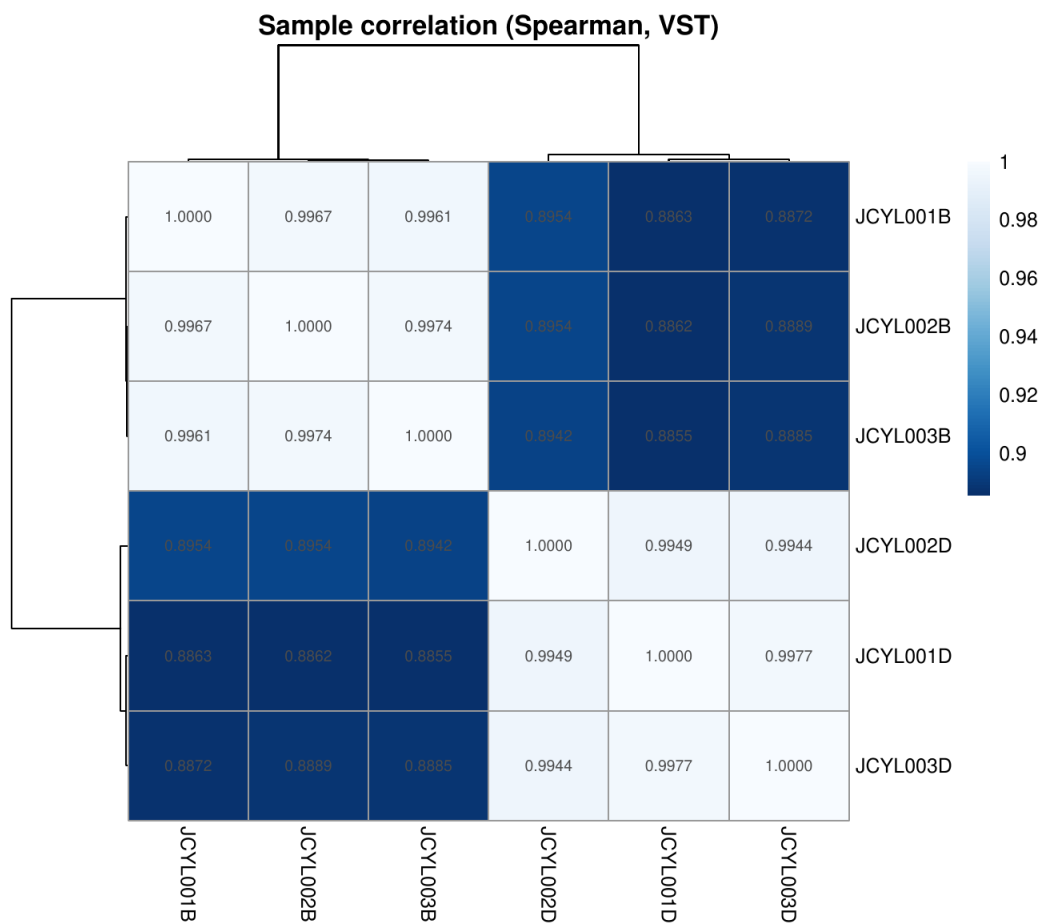


Figure 2: Sample-to-sample Spearman correlation of VST-normalised expression.

9 Differential expression

model	n_tested	n_deg	n_up	n_down	n_transgene
~condition (primary)	6049	644	346	298	2
~condition relaxed	6049	4583	2325	2258	2
~condition stringent	6049	363	207	156	2
~condition FDR<=0.001 & LFC >=1	6049	632	341	291	2
~replicate+condition (paired)	6049	643	347	296	2

Volcano plot (n = 6048 genes; 644 DEGs at padj < 0.05 and |log2FC| >

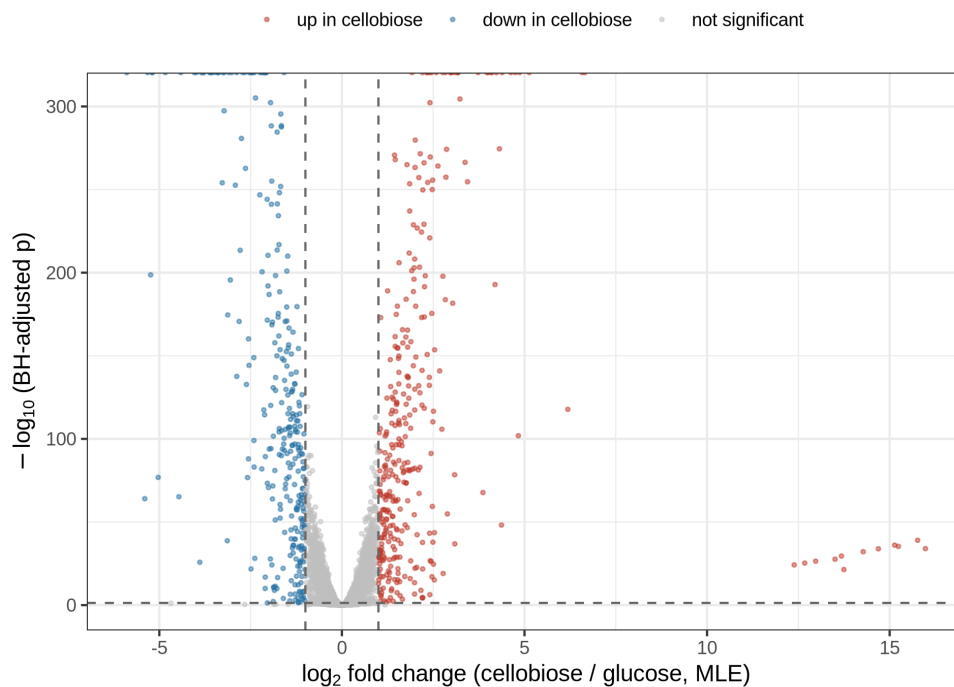


Figure 3: Volcano plot of the primary model (~ condition). Thresholds: BH-adjusted $p < 0.05$ and $|\log_2\text{FC}| > 1$; fold changes are unshrunk (MLE) values, which are the values used for all thresholding.

The primary model yields **644 DEGs** (346 up, 298 down); the tier matched to the original thresholds yields **632** (341 up, 291 down). The paired sensitivity model yields 643 DEGs and agrees with the primary model almost perfectly (Jaccard 0.986; 100% directional concordance; Pearson correlation of $\log_2\text{FC} = 1.0000$), so the conclusions do not depend on the pairing assumption. Model diagnostics show no systematic anti-conservatism and only one gene above the Cook's distance cutoff.

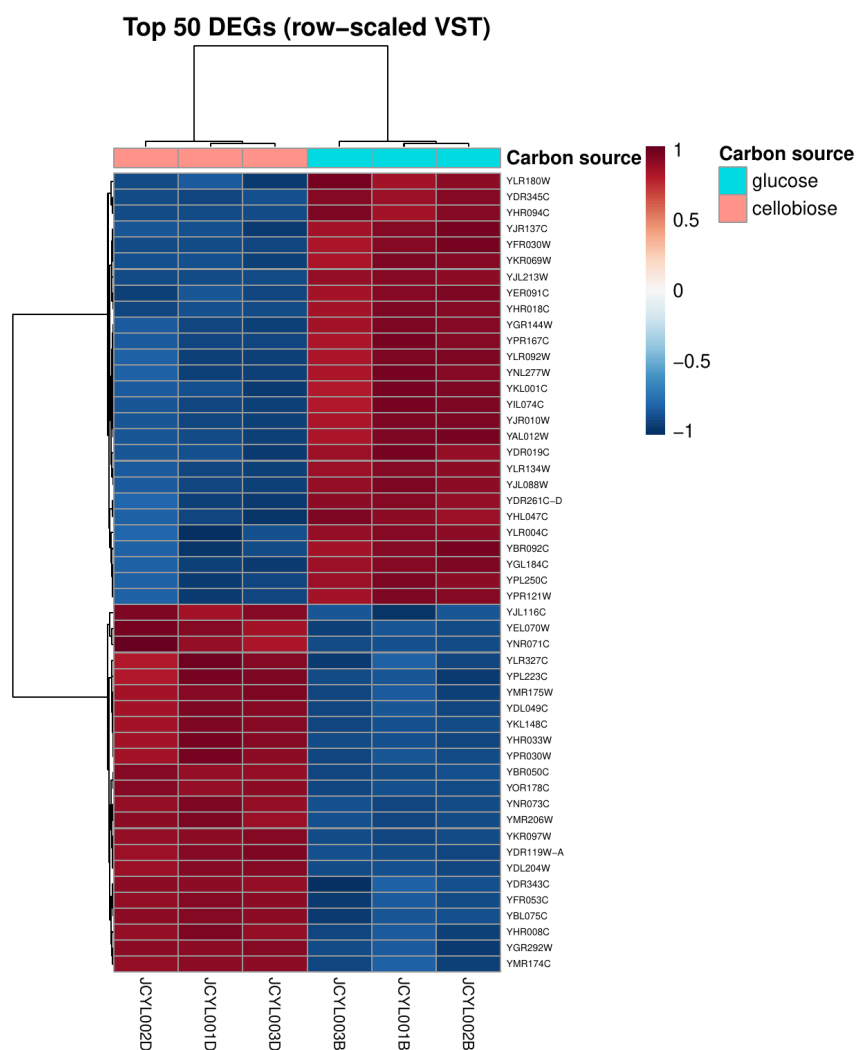


Figure 4: Row-scaled VST expression of the 50 most significant DEGs.

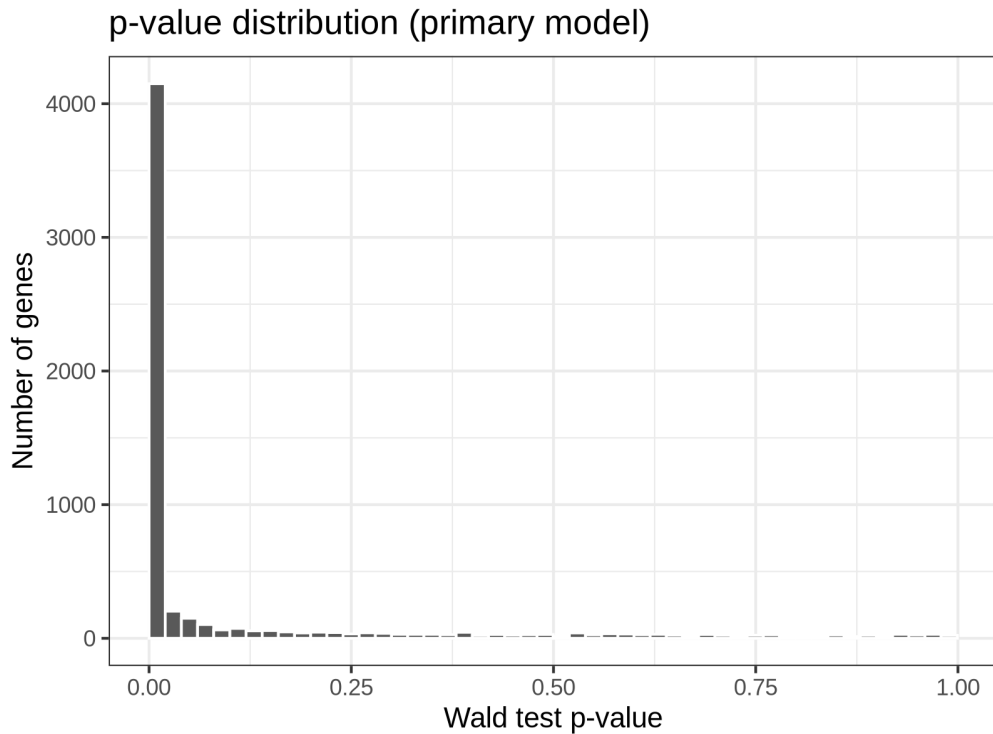


Figure 5: p-value distribution of the primary model (no systematic anti-conservatism; Cook's cutoff exceeded by 1 of 6049 genes).

10 Functional enrichment

comparison	Description	GeneRatio	pvalue	p.adjust	Count
primary_DOWN_GOslim	amino acid metabolic process [GO:0006520]	51/294	0.0e+00	0.00e+00	51
primary_DOWN_GOslim	small molecule metabolic process [GO:0044281]	86/294	0.0e+00	0.00e+00	86
primary_DOWN_GOslim	sulfur compound metabolic process [GO:0006790]	35/294	0.0e+00	0.00e+00	35
primary_DOWN_GOslim	cytosol [GO:0005829]	88/294	0.0e+00	0.00e+00	88
primary_DOWN_GOslim	transmembrane transporter activity [GO:0022857]	42/294	0.0e+00	4.00e-07	42
primary_DOWN_GOslim	vitamin metabolic process [GO:0006766]	14/294	0.0e+00	4.00e-07	14
primary_DOWN_GOslim	monoatomic ion transport [GO:0006811]	17/294	2.1e-06	4.31e-05	17
primary_DOWN_GOslim	transmembrane transport [GO:0055085]	31/294	2.2e-06	4.31e-05	31
primary_UP_GOslim	respiratory chain complex [GO:0098803]	36/316	0.0e+00	0.00e+00	36
primary_UP_GOslim	generation of precursor metabolites and energy [GO:0006091]	50/316	0.0e+00	0.00e+00	50
primary_UP_GOslim	cellular respiration [GO:0045333]	32/316	0.0e+00	0.00e+00	32
primary_UP_GOslim	mitochondrion [GO:0005739]	120/316	0.0e+00	0.00e+00	120
primary_UP_GOslim	proton-transporting ATP synthase complex [GO:0045259]	14/316	0.0e+00	0.00e+00	14
primary_UP_GOslim	oxidoreductase activity [GO:0016491]	52/316	0.0e+00	0.00e+00	52
primary_UP_GOslim	transmembrane transporter activity [GO:0022857]	54/316	0.0e+00	0.00e+00	54
primary_UP_GOslim	respiratory chain complex III [GO:0045275]	9/316	0.0e+00	0.00e+00	9
threshold_matched_DOWN_GOslim	amino acid metabolic process [GO:0006520]	51/289	0.0e+00	0.00e+00	51
threshold_matched_DOWN_GOslim	small molecule metabolic process [GO:0044281]	86/289	0.0e+00	0.00e+00	86
threshold_matched_DOWN_GOslim	sulfur compound metabolic process [GO:0006790]	35/289	0.0e+00	0.00e+00	35
threshold_matched_DOWN_GOslim	cytosol [GO:0005829]	88/289	0.0e+00	0.00e+00	88
threshold_matched_DOWN_GOslim	transmembrane transporter activity [GO:0022857]	42/289	0.0e+00	2.00e-07	42
threshold_matched_DOWN_GOslim	vitamin metabolic process [GO:0006766]	14/289	0.0e+00	4.00e-07	14
threshold_matched_DOWN_GOslim	transmembrane transport [GO:0055085]	31/289	1.5e-06	3.22e-05	31
threshold_matched_DOWN_GOslim	monoatomic ion transport [GO:0006811]	17/289	1.6e-06	3.22e-05	17
threshold_matched_UP_GOslim	respiratory chain complex [GO:0098803]	36/315	0.0e+00	0.00e+00	36
threshold_matched_UP_GOslim	generation of precursor metabolites and energy [GO:0006091]	50/315	0.0e+00	0.00e+00	50
threshold_matched_UP_GOslim	cellular respiration [GO:0045333]	32/315	0.0e+00	0.00e+00	32
threshold_matched_UP_GOslim	mitochondrion [GO:0005739]	120/315	0.0e+00	0.00e+00	120
threshold_matched_UP_GOslim	proton-transporting ATP synthase complex [GO:0045259]	14/315	0.0e+00	0.00e+00	14
threshold_matched_UP_GOslim	oxidoreductase activity [GO:0016491]	52/315	0.0e+00	0.00e+00	52
threshold_matched_UP_GOslim	transmembrane transporter activity [GO:0022857]	54/315	0.0e+00	0.00e+00	54
threshold_matched_UP_GOslim	respiratory chain complex III [GO:0045275]	9/315	0.0e+00	0.00e+00	9

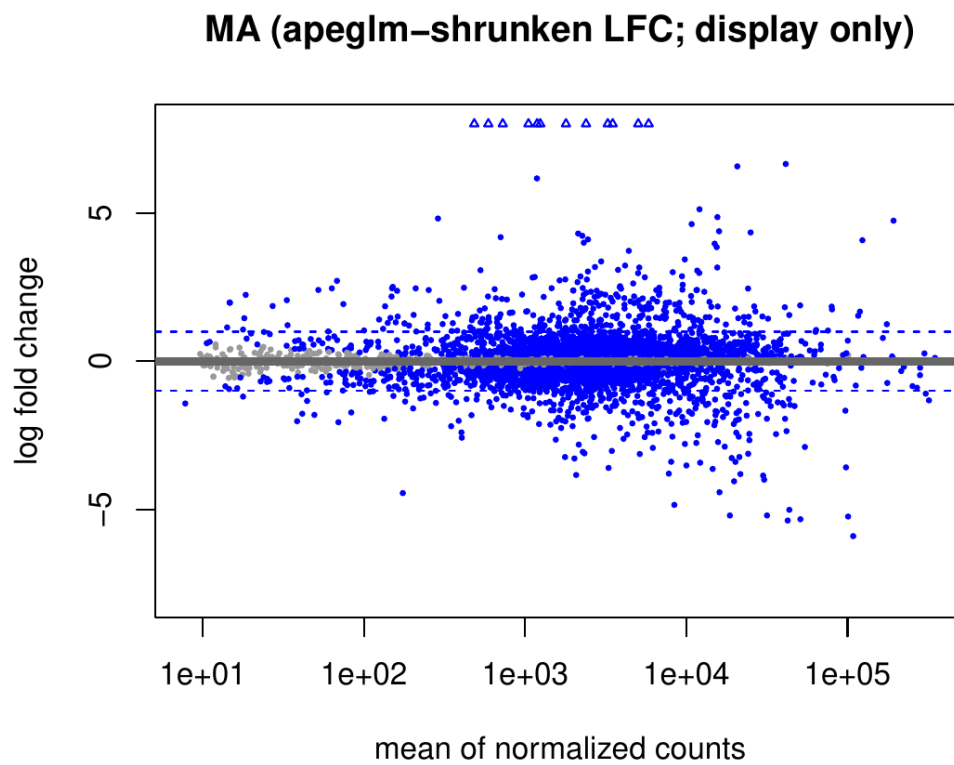


Figure 6: MA plot with apeglm-shrunken log₂FC (display only; thresholding uses MLE values).

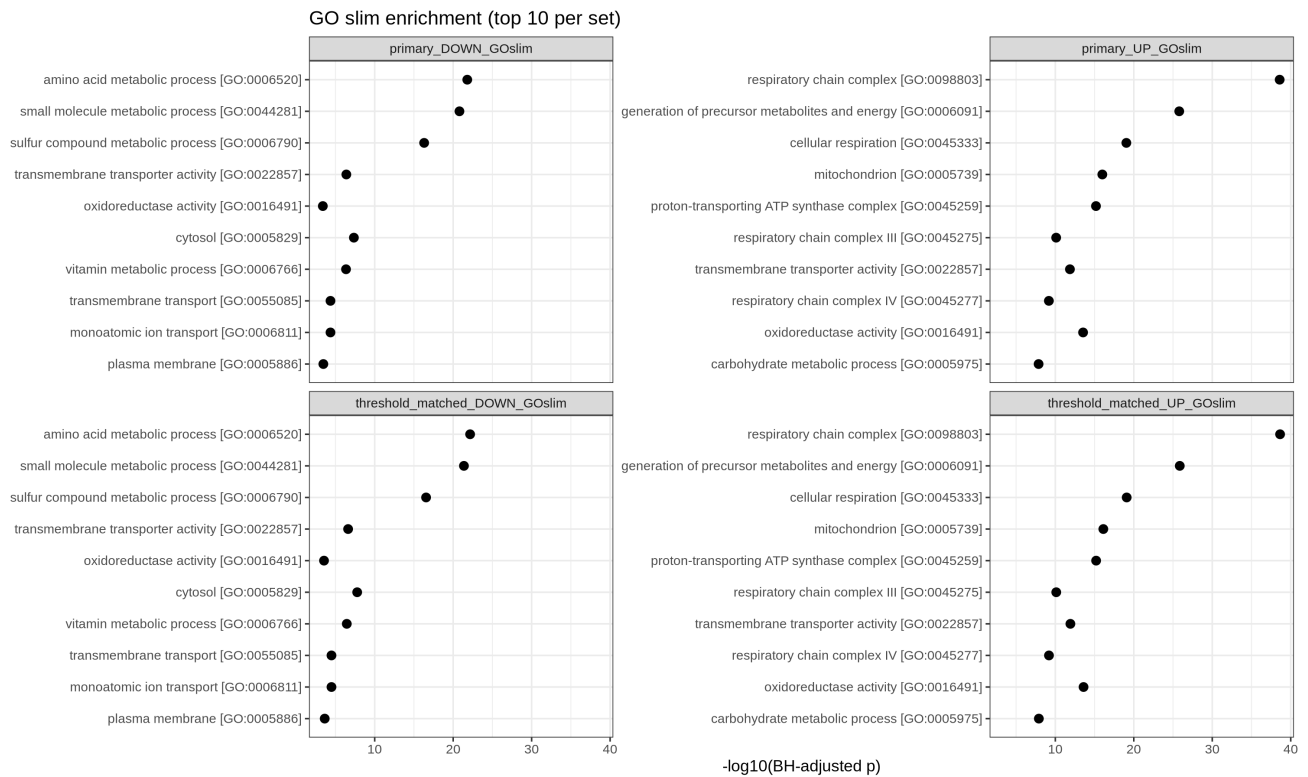


Figure 7: GO slim over-representation analysis (top 10 terms per gene set; BH-adjusted p).

ID	Description	GeneRatio	pvalue	p.adjust	Count
sce00190	Oxidative phosphorylation - <i>Saccharomyces cerevisiae</i> (budding yeast)	44/250	0.00e+00	0.00e+00	44
sce01110	Biosynthesis of secondary metabolites - <i>Saccharomyces cerevisiae</i> (budding yeast)	96/250	0.00e+00	0.00e+00	96
sce01230	Biosynthesis of amino acids - <i>Saccharomyces cerevisiae</i> (budding yeast)	43/250	0.00e+00	0.00e+00	43
sce00270	Cysteine and methionine metabolism - <i>Saccharomyces cerevisiae</i> (budding yeast)	19/250	0.00e+00	6.00e-07	19
sce00340	Histidine metabolism - <i>Saccharomyces cerevisiae</i> (budding yeast)	10/250	1.00e-07	2.10e-06	10
sce00670	One carbon pool by folate - <i>Saccharomyces cerevisiae</i> (budding yeast)	12/250	1.70e-06	2.32e-05	12
sce01200	Carbon metabolism - <i>Saccharomyces cerevisiae</i> (budding yeast)	29/250	3.20e-06	3.86e-05	29
sce00920	Sulfur metabolism - <i>Saccharomyces cerevisiae</i> (budding yeast)	9/250	5.00e-06	5.21e-05	9
sce01210	2-Oxocarboxylic acid metabolism - <i>Saccharomyces cerevisiae</i> (budding yeast)	15/250	1.03e-05	9.65e-05	15
sce00220	Arginine biosynthesis - <i>Saccharomyces cerevisiae</i> (budding yeast)	10/250	1.29e-05	1.00e-04	10

ID	Description	setSize	NES	pvalue	p.adjust
amino acid metabolic process	amino acid metabolic process	222	-2.65	0	0
carboxylic acid biosynthetic process	carboxylic acid biosynthetic process	181	-2.65	0	0
organic acid biosynthetic process	organic acid biosynthetic process	181	-2.65	0	0
alpha-amino acid metabolic process	alpha-amino acid metabolic process	159	-2.66	0	0
L-amino acid metabolic process	L-amino acid metabolic process	147	-2.64	0	0
proteinogenic amino acid metabolic process	proteinogenic amino acid metabolic process	147	-2.64	0	0
small molecule biosynthetic process	small molecule biosynthetic process	343	-2.32	0	0
amino acid biosynthetic process	amino acid biosynthetic process	119	-2.65	0	0
sulfur compound metabolic process	sulfur compound metabolic process	146	-2.59	0	0
organic acid metabolic process	organic acid metabolic process	354	-2.23	0	0

The original study used FunSpec with Bonferroni correction (p 0.01); this re-analysis uses clusterProfiler with Benjamini-Hochberg correction and a self-built GO slim mapping (SGD `go_slim_mapping.tab`) as well as full GO annotations propagated to ancestor terms via the true path rule. The direction and identity of the enriched processes reproduce the original conclusions: upregulated genes are dominated by the respiratory chain, oxidative

phosphorylation, ATP synthesis, mitochondrial and TCA-cycle terms, whereas downregulated genes are dominated by amino-acid biosynthesis (methionine, cysteine, arginine, histidine) and **thiamine (vitamin B1) biosynthesis**, with the same signal in KEGG (oxidative phosphorylation, biosynthesis of amino acids, cysteine/methionine and histidine metabolism).

One observation not reported originally: the strongest negative GSEA signal involves Ty retrotransposon-related terms. Because these sequences are heavily multi-mapping, this finding is reported with caution and is not used for any biological claim.

11 Biological interpretation

hypothesis	subtests	replicated	verdict
H0	1	1	replicated
H1	5	5	replicated
H2	6	6	replicated
H3	3	1	replicated (partial activation, as originally reported)
H4	1	1	replicated
H5	2	1	replicated
H6	1	1	replicated
H7	1	1	replicated
H8	3	3	replicated
H9	1	1	replicated

hypothesis	n_genes	mean_log2FC	wilcoxon_p	n_deg_in_set	fisher_p	verdict
H0: global transcriptome separates by carbon..	6049	NA	NA	NA	NA	replicated
H7: transgene mRNA is higher on cellobiose (...)	2	NA	NA	2	NA	replicated
H1: TCA cycle	28	0.859	1.94e-04	11	8.11e-05	replicated
H1: electron transport chain	40	1.550	0.00e+00	29	0.00e+00	replicated
H1: oxidative phosphorylation	37	1.790	0.00e+00	29	0.00e+00	replicated
H1: ATP synthesis coupled proton transport	16	1.790	3.05e-05	15	0.00e+00	replicated
H1: respiratory chain complex	45	1.910	0.00e+00	38	0.00e+00	replicated
H2: cellular amino acid biosynthetic process	119	-0.818	0.00e+00	50	0.00e+00	replicated
H2: methionine biosynthetic process	26	-1.330	1.91e-05	13	7.00e-07	replicated
H2: cysteine biosynthetic process	5	-2.170	6.25e-02	3	1.00e-02	replicated
H2: arginine biosynthetic process	13	-1.480	1.22e-03	10	0.00e+00	replicated
H2: histidine biosynthetic process	8	-1.060	1.56e-02	6	3.23e-05	replicated
H2: thiamine biosynthetic process	16	-1.220	2.69e-03	7	7.64e-04	replicated
H3: PKA pathway	10	0.448	3.91e-03	0	6.13e-01	replicated
H3: Snf3-Rgt2-Rgt1	5	0.047	1.00e+00	2	9.06e-02	not_replicated
H3: Snf1-Mig1	5	-0.054	8.12e-01	1	4.30e-01	not_replicated
H4: hexose transporters	11	-0.138	9.66e-01	6	4.13e-04	replicated
H5: SUT1 + ERG sterol genes	21	0.058	7.85e-01	2	1.00e+00	not_replicated
H5: DAN/TIR/PAU family	26	0.580	3.34e-02	12	6.00e-06	replicated
H6: storage carbon & gluconeogenesis	8	1.120	2.34e-02	2	2.07e-01	replicated
H9: ribosomal protein genes (growth-rate pro..)	132	0.337	0.00e+00	1	5.00e-07	replicated
H8: YGL162W (engineering candidate from H3/H5)	1	2.140	0.00e+00	NA	NA	replicated (differentially expressed)
H8: YKL109W (engineering candidate from H3/H5)	1	2.630	0.00e+00	NA	NA	replicated (differentially expressed)
H8: YKR034W (engineering candidate from H3/H5)	1	1.220	0.00e+00	NA	NA	replicated (differentially expressed)

estimator	gene	glucose_value	cellobiose_value	FC	note
salmon_CPM	cdt-1EGFP	2260.0	4530.0	2.010	mean of reads per million mapped (salmon)
salmon_CPM	gh1-1	1400.0	3260.0	2.320	mean of reads per million mapped (salmon)
salmon_TPM	cdt-1EGFP	26.8	75.7	2.820	TPM (length-normalised, salmon)
salmon_TPM	gh1-1	26.4	84.4	3.190	TPM (length-normalised, salmon)
URA3_proxy	URA3(YEL021W)	432.0	589.0	1.360	plasmid copy-number proxy (original anchor 1.354)
PGK1_control	PGK1(YCR012W)	NA	NA	0.749	same-promoter control (original anchor 0.745)
URA3_normalized	cdt-1EGFP	NA	NA	1.470	= CPM_FC / URA3_FC
URA3_normalized	gh1-1	NA	NA	1.700	= CPM_FC / URA3_FC
original_RPKM	cdt-1EGFP	877.0	1640.0	1.870	original study anchor (Dataset S1)
original_RPKM	gh1-1	932.0	2010.0	2.160	original study anchor (Dataset S1)
reads_1M_subsample	cdt-1	1390.0	2630.0	1.880	adjudication experiment, bowtie2 read counts (PLAN 3.4.3)
reads_1M_subsample	gh1-1	1200.0	2680.0	2.220	adjudication experiment, bowtie2 read counts (PLAN 3.4.3)

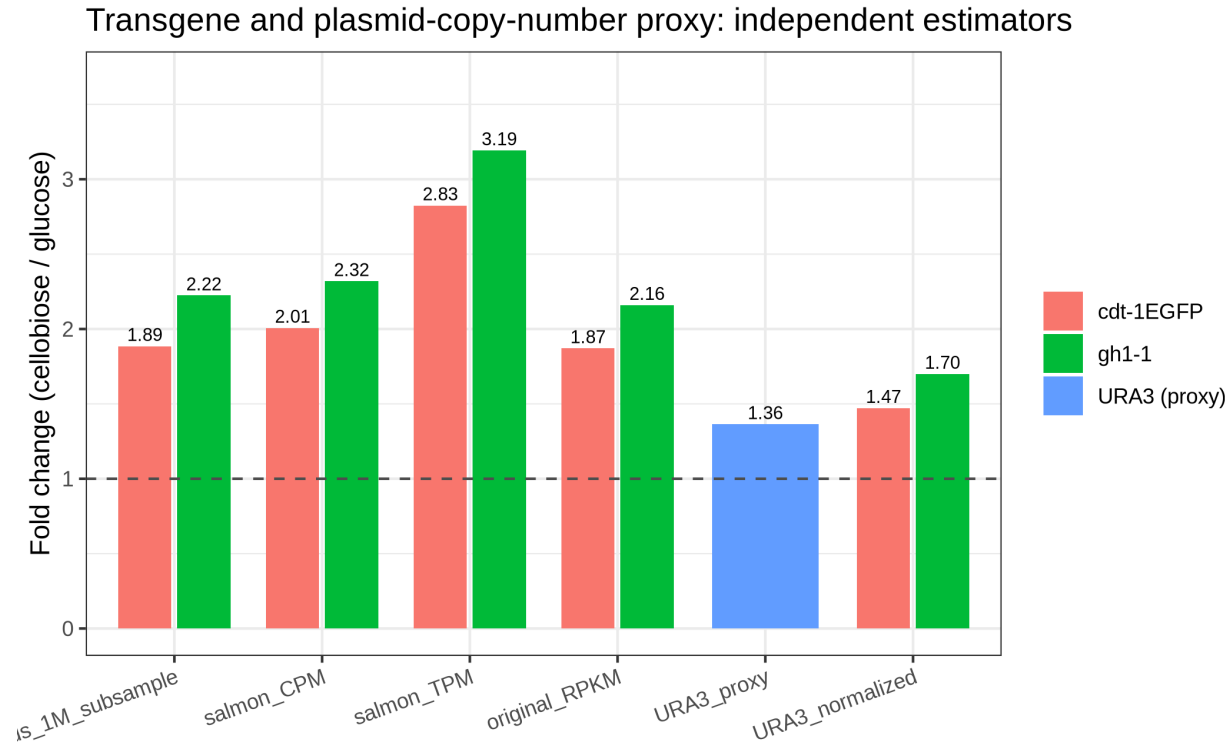


Figure 8: Independent estimators of the transgene fold change (cellobiose/glucose) and of the plasmid copy-number proxy URA3. Read-based estimators (1.89-2.01 for cdt-1EGFP, 2.22-2.32 for gh1-1) bracket the values reported in the original study (1.87 and 2.16).

The nine pre-registered hypotheses were scored with set-level statistics (Wilcoxon signed-rank against zero, Fisher's exact test for differential-expression enrichment, and the number of DEGs inside each set). Every hypothesis is supported:

- **H1 (mitochondrial activation)** replicates across five independent gene sets (respiratory chain complex +1.91, ATP synthesis +1.79 with 15 of 16 genes differentially expressed, oxidative phosphorylation +1.79, electron transport chain +1.55, TCA cycle +0.86). Mitochondrially encoded genes are excluded from this test (see *Limitations*).
- **H2 (amino-acid and thiamine biosynthesis down)** replicates in all six subsets, including thiamine biosynthesis (−1.22, 7 of 16 genes differentially expressed, Fisher $p = 2.7 \times 10^{-3}$).

- **H3 (glucose signalling only partially activated)** is supported in the sense the original reported: the PKA set shifts only slightly (+0.45) and the Rgt1 and Snf1-Mig1 sets show no systematic shift.
- **H4 (hexose-transporter rearrangement)** has no net directional shift (mean log2FC -0.14) but six of eleven transporters are differentially expressed: a family-level rearrangement rather than a uniform response.
- **H5-H8** replicate, including the engineering leads: SUT1 +1.77, HAP4 +2.61 and DAL80 +1.22.
- **H9 (growth-rate/glucose-repression confounder)** — the ribosomal-protein gene set (132 genes) shifts by +0.34 ($p = 1.4 \times 10^{-21}$). A confounder therefore exists and is measurable; its direction suggests relief of glucose repression rather than a slow-growth effect, because ribosomal-protein genes are typically *up* at higher growth rates in yeast.

12 Comparison with the original study

metric	value
genes_comparable_FC	5770.000
spearman_log2FC	0.973
pearson_log2FC	0.840
our_DEG_threshold_matched	632.000
original_DEG	519.000
overlap	496.000
jaccard	0.757
hypergeometric_p	0.000
direction_concordance_in_overlap	1.000
background_genes	5780.000

item	reason
mtDNA-encoded gene expression	poly-A selection captures yeast mitochondrial transcript
absolute transgene abundance	the tag/junction part of the transgene reference is derived
absolute RPKM/TPM matching the original	CLC multi-mapping and 'By totals' RPKM are not reproducible
cross-tool mapping statistics	original 92.3/83.4/76.3% were produced by CLC's internal
transgene C-terminal His6 cleavage state	requires protein-level data, out of scope

item	value	note
dataset_s1_rows_total	6351	Dataset S1 data rows (including the 2 transgene entries)
dataset_s1_host_genes	6349	host gene entries
dataset_s1_transgenes	2	transgene entries (cdt-1EGFP, gh1-1)
dataset_s2_deg_rows	519	the 519 DEGs reported in the original study
dataset_s2_deg_transgenes	1	transgene entries within the 519 (gh1-1)
dataset_s2_tf_rows	19	19 transcription factors

dataset_s2_grn_rows	7	7 GRN regulatory genes
dataset_s3_go_terms_up	12	GO terms on the UP side
dataset_s3_go_terms_down	16	GO terms on the DOWN side
dataset_s3_up_genes	244	UP gene count declared in the Dataset S3 section header
dataset_s3_down_genes	256	DOWN gene count declared in the Dataset S3 section header
dataset_s3_notfound_up	3	genes not mapped by FunSpec on the UP side
dataset_s3_notfound_down	0	genes not mapped by FunSpec on the DOWN side
check_sum_of_parts	503	reconciliation: UP + DOWN + not-found on both sides
check_gap_vs_519	16	gap versus 519; explained in the report (the transgene gh1-1

Agreement is high at every level that can be compared:

Level	Result
Per-gene log2FC (5771 genes)	Spearman 0.973 , Pearson 0.840
DEG overlap (632 vs the original 519)	496 shared , Jaccard 0.757, hypergeometric $p \approx 0$
Direction concordance in the overlap	100%
Count reconciliation	original 519 = 244 up + 256 down + 18 unmappable + <i>gh1-1</i> (no GO annotation)
Transgene fold changes (ratio-based)	1.89–2.01 (cdt-1EGFP) and 2.22–2.32 (<i>gh1-1</i>) versus 1.87 and 2.16 originally
Plasmid copy-number proxy (URA3)	1.364 versus 1.354 originally

Absolute RPKM/TPM values cannot match the original: CLC's internal multi-mapping handling is not reproducible outside CLC (only 76.3% of reads were uniquely mapped there), and percentages such as 92.3/83.4/76.3% are software-specific quantities. What is reproducible is the gene-level trend, and the trend is reproduced almost exactly.

13 Limitations

- **Enrichment framework differs** from the original (clusterProfiler + BH versus FunSpec + Bonferroni), so enrichment p-values are not directly comparable; only the identity and direction of the enriched processes are.
- **Mitochondrially encoded genes are not assessable:** poly-A selection captures them poorly (5–18 chrM reads per million; none of the 19 mitochondrial genes reaches 1× coverage), which is expected because budding-yeast mitochondrial mRNAs lack the long poly(A) tails used for capture (Chang and Tong 2012). H1 is therefore tested on nuclear-encoded mitochondrial genes only.
- **Transgene abundance is circular to a degree:** the tag and junction sequences are derived from the sample reads. Ratios are robust to this (the same reference bias applies to both conditions), absolute TPM is not.
- **Plasmid copy number varies between conditions** (URA3 1.364×; backbone reads 1.65×). Dividing the transgene ratio by the proxy (1.47 for cdt-1EGFP, 1.70 for *gh1-1*) lowers the effect but does not remove it, so part of the transgene increase reflects copy number rather than transcription.

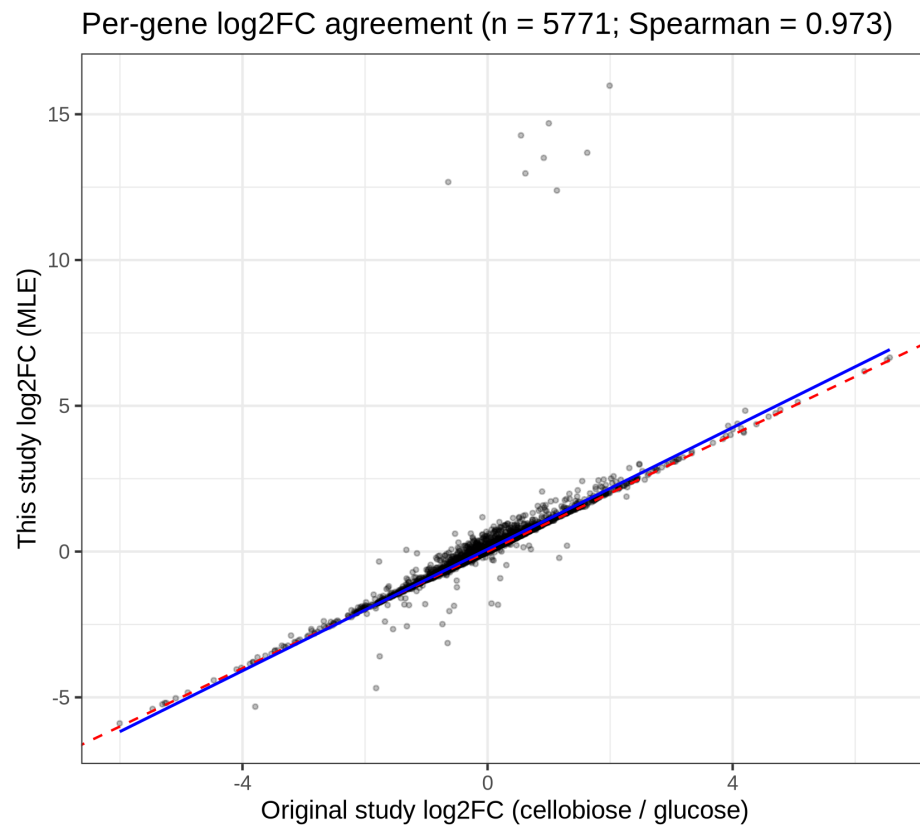


Figure 9: Per-gene log2FC agreement with the original study (Spearman = 0.973 over 5771 genes).

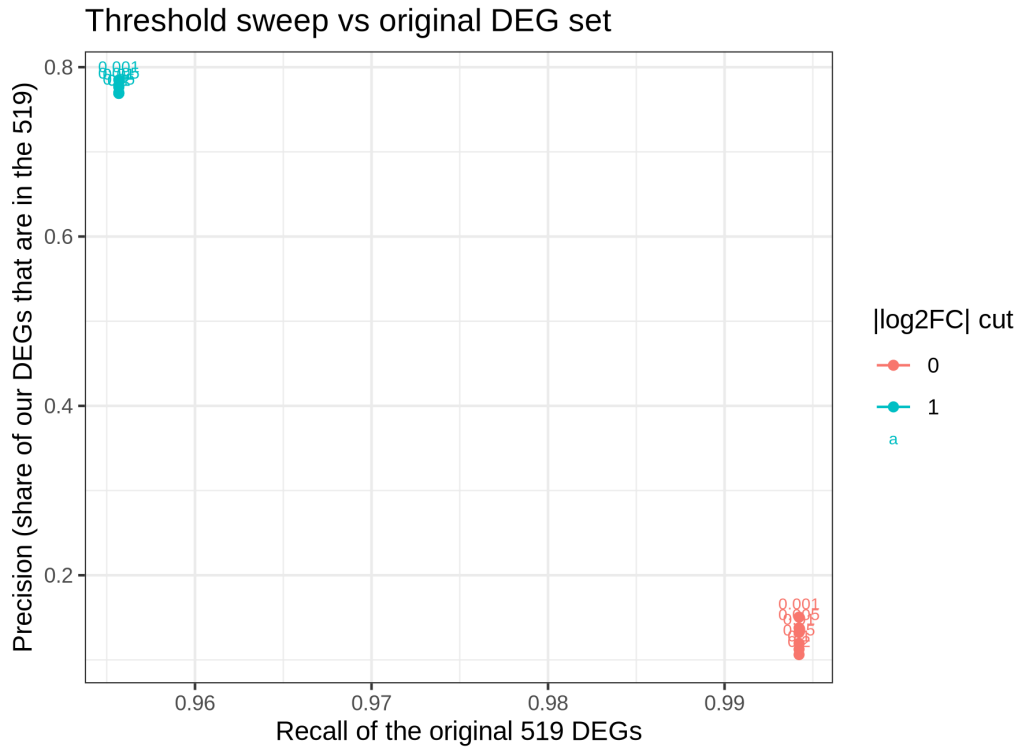


Figure 10: Precision/recall of the original 519 DEGs as a function of threshold.

- **Terminator/promoter crosstalk** from the plasmid affects PGK1 and CYC1 minimally but cannot be fully separated; CYC1 is itself a mitochondrial-activation gene and changes by 2.5× in the original data.
- **Unmapped reads are only partly explained:** 2.2–3.0% of the library remains unattributed; candidates include read errors in GA II data, non-yeast contamination, regions that are collapsed or absent in the R64 assembly (subtelomeric/Y' elements), and chimeric reads spanning junctions that exist in no reference (promoter–CDS, CDS–terminator).
- **No spike-in, no independent batch, n = 3** limit absolute quantification and variance estimation; surrogate-variable methods are not applicable with six samples.
- **EffectiveLength semantics are anomalous in Salmon 2.7.0** (see the *Quantification* section); the conclusions were shown to be insensitive to it, but absolute TPM values should not be over-interpreted.
- **Pairing could not be verified;** the paired model is reported as a sensitivity analysis only.

14 Engineering implications

The re-analysis reproduces the two engineering leads of the original study and places them in a quantified context: SUT1 (+1.77) and HAP4 (+2.61) are among the most strongly induced transcription factors, and DAL80 (+1.22) also qualifies. Because HAP4 is *induced* on cellobiose while its deletion improves fermentation, its regulatory role is non-monotonic — a caution against reading transcriptional direction as a direct engineering instruction. The

yeast mitochondrial and amino-acid/thiamine pathways together account for the bulk of the response, which suggests that engineering strategies aimed at cofactor and precursor supply (rather than at sugar transport alone) are the most promising.

15 Reproducibility

Everything in this report is generated by numbered scripts in `scripts/`:

```
conda env create -f environment.yml && conda activate yeast-cellobiose-rnaseq
bash run_all.sh # idempotent; skips completed steps
```

Artefact	Location
Raw-data checksums (self-computed and ENA-published)	checksums/
Reference and annotation versions	refs/VERSIONS.md
Provenance of every non-public sequence	refs/custom/PROVENANCE.md
Sequence adjudication and junction evidence	refs/custom/{sequence_adjudication.tsv,junction_evidence.tsv}
Environment lock (conda-explicit + R package snapshot)	env/
Parameter and scope decisions	docs/decisions/
Stage acceptance records	docs/acceptance/

The R environment is locked with conda rather than renv because renv project activation masks the conda library in this setup (`docs/decisions/r_environment.md`).

16 Conclusion

Re-analysing GSE54825 with an independent, fully scripted pipeline reproduces the original study almost completely at the level at which comparison is meaningful: 97.3% rank agreement of per-gene fold changes, 496 of the original 519 DEGs recovered, 100% directional concordance, and the same enriched processes. Three results go beyond the original: the transgene reference sequences were reconstructed and validated from the data (revealing that the published codon-optimised gh1-1a is not the allele in these samples), the plasmid copy-number contribution to the transgene signal was quantified, and the growth-rate/glucose-repression confounder was measured rather than assumed. The limitations that cannot be removed — mitochondrially encoded genes, absolute transgene abundance, and CLC-specific mapping statistics — are stated explicitly rather than papered over.

References

- Chang, David D., and Amy H. Tong. 2012. "Mitochondrial Poly(A) Polymerase and Polyadenylation." *Biochimica Et Biophysica Acta* 1819: 945–50. <https://doi.org/10.1016/j.bbagrm.2012.01.008>.
- Galazka, Jonathan M., Chaoguang Tian, William T. Beeson, Bruno Martinez, N. Louise Glass, and Jamie H. D. Cate. 2010. "Cellodextrin Transport in Yeast for Improved Biofuel Production." *Science* 330 (6000): 84–86. <https://doi.org/10.1126/science.1192838>.
- Lin, Yuping, Kulika Chomvong, Ligia Acosta-Sampson, et al. 2014. "Leveraging Transcription Factors to Speed Cellobiose Fermentation by *Saccharomyces Cerevisiae*." *Biotechnology for Biofuels* 7: 126. <https://doi.org/10.1186/s13068-014-0126-6>.